

IMPERIAL

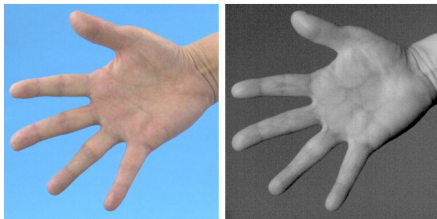
Edge-Deployable NIR-to-RGB Translation

A Pre-Processor for Pre-Trained Vision Models

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25 June 2026

Why near-infrared?

NIR sees what visible light misses, and that is what RGB models get wrong



Skin (RGB left, NIR right): NIR penetrates, exposing subsurface veins.

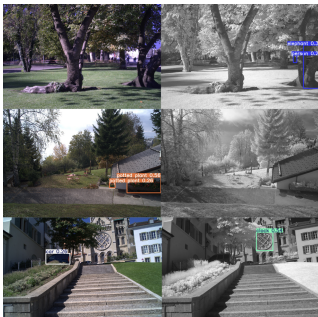


Trees (RGB left, NIR right): foliage glows in NIR because chlorophyll reflects it.

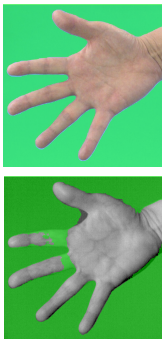
- Penetrates skin, vegetation reflects, water absorbs (biometrics, agriculture, remote sensing).
- Sees in darkness and fog (night vision). **But** it has no colour and materials look different, so RGB models misread it.

The problem: RGB models break on NIR

No colour, different materials: detectors and segmentation collapse



Detection: YOLOv8



Segmentation: MediaPipe (top RGB, bottom NIR)



Segmentation: BEiT

In each: **RGB** (works) vs. the matched **NIR** (fails). Retraining every model on labelled NIR is impractical.

The idea

Insert a learned translator and leave the downstream model untouched



Key message: judge a translation by how much **frozen-model behaviour** it recovers, not by how good the image looks.

Why it is hard, and the two design choices that follow

The task

- No colour in NIR, so it is **one-to-many**: many RGBs fit one frame.
- Materials look different (foliage bright, water dark).
- Aligned pairs are scarce (camera parallax).

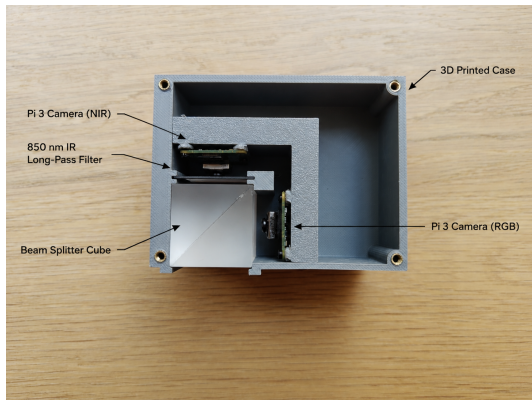
Prior work

- GANs and diffusion, judged mainly on **image quality**.
- Diffusion is iterative and stochastic, wrong for a deterministic edge pre-processor.

Two design choices: (1) a deterministic, edge-friendly **restoration backbone**; (2) supervise in **feature space**, not pixels.

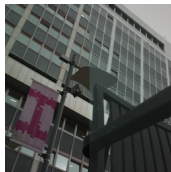
Data: a shared-aperture capture rig

12,536 simultaneous, homography-aligned NIR/RGB pairs across London



Raspberry Pi 5 with two Camera Module 3 sensors, one behind an 850 nm long-pass filter.

- A **beam-splitter cube** gives both cameras one optical centre, so a single homography aligns the whole frame.
- 20+ hours cycling London, filtered and aligned to **12,536** pairs at 256×256 .
- No public NIR/RGB set is captured and aligned simultaneously at this scale.



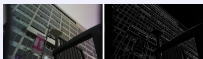
An aligned pair (half RGB / half NIR).

Producing aligned pairs

Shared aperture means one depth-independent homography per pair



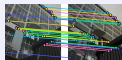
1. Raw pair: full-frame RGB/NIR straight from the two sensors.



2. Sharpness filter: reject motion-blurred frames (Laplacian variance < 100).



3. Centre-crop: common square field of view for the feature detector.



4. SIFT + RANSAC: match keypoints, fit the NIR $\rightarrow$ RGB homography.



5. Warp + resize: aligned 256×256 pair, degenerate fits discarded.

The translator: an edge-friendly backbone, a feature-space loss

An activation-free NAFNet, trained to match frozen foundation features

Backbone: activation-free

No ReLU or GELU. The only nonlinearity is the

SimpleGate: split the channels in half and multiply the two halves. **First use of NAFNet for NIR→RGB.**

Why this beats a ReLU:

- Keeps negatives, so no “dead” units and steadier training.
- Gates one half with the other, capturing feature interactions.
- Predictable value range, so it quantises to INT8 cleanly.
- Small, standard op set, so clean export to ONNX, LiteRT, Hailo.

Loss: supervise in feature space

Replace the single VGG perceptual term with a **frozen foundation ensemble**, used as a training signal on an unmodified backbone:

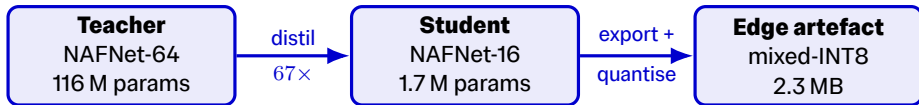
- **DINOv2**: self-supervised features
- **CLIP**: vision-language features
- **ResNet-50**: ImageNet features

(kept alongside L1, MS-SSIM and a light PatchGAN.)

Train so that, for each frozen extractor F:

$$F(T(x_{\text{NIR}})) \approx F(x_{\text{RGB}})$$

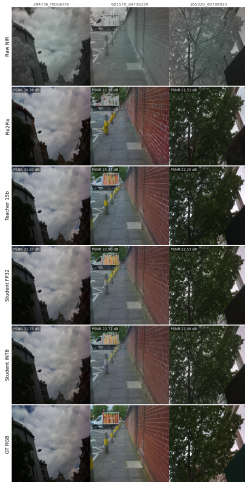
Making it deployable: distil, then quantise



- Student inherits the teacher's activation-free blocks $\Rightarrow$ same clean operator set.
- Distilled on the teacher's outputs + frozen-feature losses (downstream fidelity, not pixel mimicry).
- Post-training INT8, exported to LiteRT (CPU) and a Hailo-8L HEF (NPU).

Does it actually work?

NIR in, RGB out: the eye test is not enough



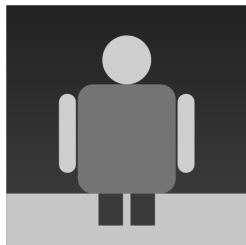
Rows, top to bottom

1. Raw NIR (input)
2. Pix2Pix baseline
3. Teacher (NAFNet-64)
4. Student FP32
5. Student INT8 (deployed)
6. Ground-truth RGB

All produce plausible colour, and Pix2Pix even ties the student on PSNR and SSIM. But it recovers almost **none** of the downstream behaviour.

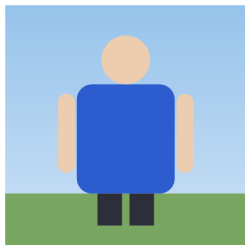
Why not just report PSNR?

One NIR input has many RGB translations, and PSNR picks the wrong one



NIR input

what the translator sees



Translation A

same person, blue shirt

PSNR 13.0 dB



Translation B

a red postbox

PSNR 18.2 dB



True RGB

the target (GT)

Both are valid RGB guesses for one NIR input. Against the true RGB, PSNR scores the **postbox** higher than the real person, just for keeping the red pixels. It rewards pixels, not meaning.

How I measure success: gap closure

Run a frozen model on three inputs and ask how much RGB behaviour is recovered

$$\text{gap closure} = \frac{m(\text{translated}) - m(\text{NIR})}{m(\text{RGB}) - m(\text{NIR})}$$

0

no better than raw NIR

1

native-RGB behaviour fully
recovered

< 0

translation made it worse

Reference = the same frozen model on native RGB (pseudo-labels), across detection, segmentation, depth and independent embedding probes. No human annotation required.

Headline result

Same pixel metrics, opposite downstream story

Pixel fidelity	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
Pix2Pix	24.03	0.752	0.185
Student FP32	24.09	0.781	0.146
Student INT8	23.57	0.770	0.153

Comparable on pixels. Now the downstream gap closure:

Task (gap closure)	Pix2Pix	Student FP32	Student INT8
Detection (YOLOv8m)	-0.15	+0.08	+0.08
Detection (RT-DETR-L)	-0.13	+0.07	+0.05
Segmentation (ADE20K)	+0.18	+0.59	+0.57
Segmentation (Cityscapes)	+0.23	+0.35	+0.31
Depth (DepthAnything)	-0.00	+0.37	+0.33
(indep.) DINOv2-L	-0.02	+0.57	+0.53
(indep.) SigLIP2	+0.10	+0.54	+0.51

Pix2Pix **matches the student on pixel metrics** but goes negative on both detectors and recovers almost nothing downstream. Both student precisions stay positive on all five primary tasks.

What drives it: the feature ensemble

Remove it: best pixel scores, yet it trails on every downstream metric

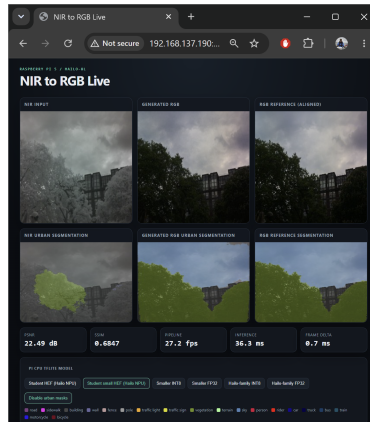
Teacher config	pixel		downstream			
	PSNR $\uparrow$	LPIPS $\downarrow$	DINOv2-L $\uparrow$	YOLO clo. $\uparrow$	ADE clo. $\uparrow$	Depth $\downarrow$
Pix2Pix baseline	24.03	0.185	0.723	-0.152	0.183	0.678
No feature-match	27.10	0.098	0.886	+0.011	0.619	0.420
Final teacher	26.16	0.103	0.921	+0.108	0.670	0.365

Without the ensemble the teacher gets the **best PSNR and LPIPS**, yet the final teacher leads on **every downstream metric** and detection closure collapses to near zero. The ensemble, not pixel fidelity, recovers downstream behaviour.

It runs on the edge, in real time

Backend	FPS	Size
Hailo-8L NPU	27.2	17.7 MB
CPU LiteRT INT8	5.05	2.3 MB
CPU LiteRT FP32	4.40	7.1 MB

- Meets the **5 FPS** target on CPU and clears it 5.4× on the NPU.
- Peak memory <50 MB, no swap, no thermal throttling.
- INT8 is faster than FP32 here, from narrower kernels and less bandwidth.



► **Live demo (switch to video)**

Live on-device web app: NIR, translated RGB and a live RGB reference segmented side by side on the Pi.

Limitations

- **Behaviour, not accuracy.** Reference outputs are the frozen models' own RGB predictions (pseudo-labels), not human annotations.
- **People untested.** Privacy-by-design excluded close-range faces, and person is already among the weakest segmentation classes.
- **Geographically narrow.** Outdoor London only, one rig and one sensor.
- **Trained on sharp frames only.** Blurred and motion-corrupted captures were filtered out, so quality degrades on blurry or noisy input.

Conclusion

Pixel fidelity does **not** predict downstream usefulness. What recovers behaviour is training and evaluating in the models' own feature spaces.

Three contributions, new in combination:

1. A **shared-aperture rig** giving **12,536** simultaneously-aligned NIR/RGB pairs, which no public dataset offers.
2. The **first use of NAFNet for NIR→RGB**, with a **foundation-feature ensemble** replacing the perceptual loss.
3. Distilled to a **1.7 M** student ($67\times$ smaller) running real-time on the Pi: **5.05 FPS** CPU, **27.2 FPS** NPU.

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**Thank you.
Questions?**

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Backup slides

Backup: image-quality metrics

Why these alone would have misled

Model	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	Params
Raw NIR	16.65	0.656	0.337	n/a
Pix2Pix	24.03	0.752	0.185	54.4 M
Teacher	26.16	0.818	0.103	116 M
Student FP32	24.09	0.781	0.146	1.7 M
Student INT8	23.57	0.770	0.153	1.7 M

Pix2Pix and the student are level on pixel metrics. The downstream split (previous table) is the deciding evidence.

Backup: embedding agreement (all probes)

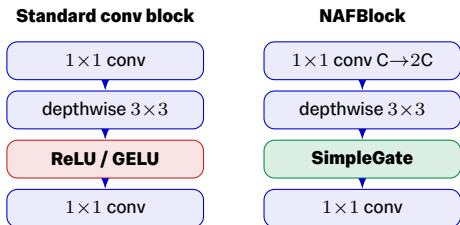
Including the training-overlapping probes the headline omits

Evaluator	Train overlap	Raw	Student FP32	FP32 closure	INT8 closure
DINOv2-S	Yes	0.727	0.923	+0.72	+0.69
CLIP-B/16	Partial	0.856	0.929	+0.50	+0.44
DINOv2-L	No	0.728	0.882	+0.57	+0.53
SigLIP2	No	0.883	0.946	+0.54	+0.51

Closure = $(\text{translated} - \text{NIR}) / (1 - \text{NIR})$ on cosine. The two overlapping probes (top) score at least as high as the independent ones. That is consistent with recovery but is **not** independent evidence, so the headline reports only DINOv2-L and SigLIP2. Distribution-level CLIP-FID: 17.86 $\rightarrow$ 5.50 (FP32) $\rightarrow$ 6.95 (INT8), lower is better.

Backup: why NAFNet?

Activation-free blocks $\Rightarrow$ predictable ranges $\Rightarrow$ clean INT8 / edge export



SimpleGate: split channels in half, **multiply** the two halves. No nonlinear activation.

Why it matters here

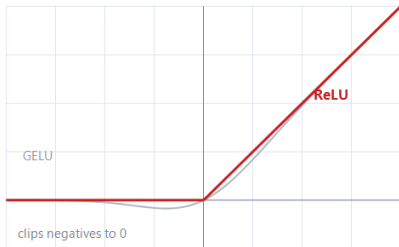
- No saturating activations $\Rightarrow$ **predictable dynamic range**, nothing to clip-calibrate for INT8.
- Small, standard operator set (conv, LayerNorm, multiply, PixelShuffle) $\Rightarrow$ **clean ONNX / LiteRT / Hailo** export.
- Zero-init residual scales $\Rightarrow$ stable training at batch size 1.

Non-standard choice: restoration backbones target same-modality tasks, but here it infers missing colour.

Backup: ReLU vs SimpleGate

The activation NAFNet removes, and the gate it uses instead

ReLU: a fixed clipping activation



Bends at zero and throws away every negative value.

SimpleGate: split the channels, multiply



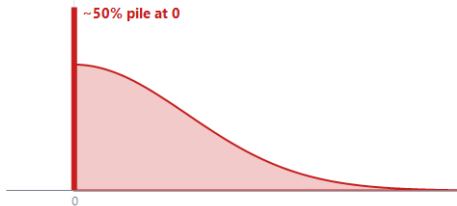
$A \times B$: still nonlinear, two-sided, nothing clipped. No activation function at all.

Same nonlinearity, no clipping. The value range stays predictable, which is what lets the network quantise to INT8 and export to the edge cleanly.

Backup: value distributions

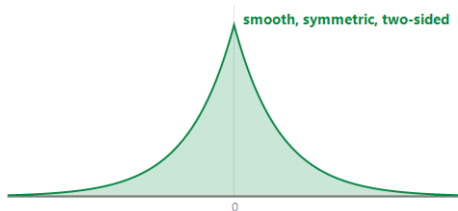
The activation choice shapes how the values are spread, which is what INT8 has to capture

ReLU output



Half the values clipped to 0, one-sided. The range is skewed and shifts during training.

SimpleGate output

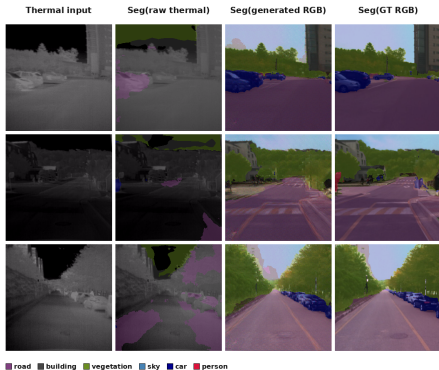


Smooth, symmetric, centred on 0. A stable, predictable range.

A fixed INT8 scale fits the smooth, symmetric range well, but wastes resolution on ReLU's spike and one-sided tail. That is why the activation-free network quantises with little error.

Backup: where the recipe generalises (and breaks)

Tested on other modalities to find the boundary of the approach



Thermal (KAIST): structures become segmentable in the translated output.

Thermal: transfers

- Recipe applied unmodified. CLIP cosine $0.46 \rightarrow 0.92$.
- Roads, vegetation and sky recover. Person still fails (heat contrast lost).

SAR: breaks down

- Pixel metrics collapse (PSNR 13.3, SSIM 0.12), only marginal downstream gains.

Boundary: works where the modality shares spectral structure with RGB, not as a universal cross-modal adapter.

Backup: anticipated questions

- **Pseudo-labels circular?** Independent probes (DINOv2-L, SigLIP2) were held out of training and still close > 0.5 .
- **Why detection weaker?** Small objects and fine texture reproduce least faithfully. Segmentation and depth lean on large-scale structure.
- **CLIP/DINOv2-S contaminated?** Yes, partially, so they are reported as supporting, not primary, evidence.
- **Why not diffusion?** Iterative and stochastic, so it violates the latency budget and determinism (F3).
- **Why not fine-tune the downstream models?** Needs weights and labelled NIR per task, out of scope. The translator leaves them unchanged.